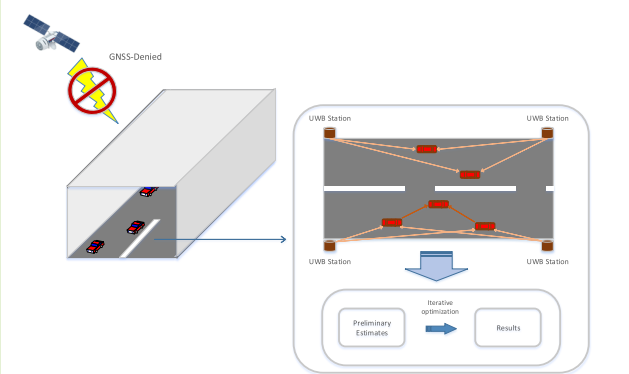
Vehicle Cooperative Localization Based on UWB Technology in GNSS-Denied Environments

Hualin Yao, Xiaomei Qu, Liangkun Wu, and Yirong Wu

***Abstract*—The accurate vehicle positioning is critical for many intelligent transportation systems (ITSs) applications. However, Global Navigation Satellite Systems (GNSS) often face significant challenges in various demanding scenarios. This article explores the cooperative vehicle localization in GNSS-denied environments by utilizing a fusion of relative distance and angle-of-arrival (AoA) information gathered from vehicles and roadside ultrawideband (UWB) base stations with known coordinates. In dense traffic situations, UWB base stations may experience nonline-of-sight (NLOS) effects when localizing vehicles. To tackle this issue, we propose a strategy that begins by classifying vehicles into line-of-sight (LOS) and NLOS categories based on distance and AoA measurements, as well as discrepancies observed between two different localization methods. Initially, we estimate the positions of LOS vehicles, which are then used to determine the positions of NLOS vehicles through the distance information connecting LOS and NLOS vehicles. Once the preliminary positions of all vehicles are established, we employ an iterative optimization algorithm to refine these estimates, ultimately yielding more accurate localization outcomes. Simulation experiments indicate that the proposed method demonstrates the excellent localization performance in both static and dynamic positioning scenarios.**



***Index Terms*— Cooperative localization, Global Navigation Satellite Systems (GNSS)-denied, nonline of sight (NLOS), ultrawideband (UWB).**

I. Introduction

IN RECENT years, intelligent transportation systems (ITSs) are increasingly being considered to alleviate traffic safety and congestion issues encountered in vehicle transportation, and vehicle positioning information forms the foundation of many ITS applications, such as navigation, location-based services, and fleet management [1]. The Global Navigation Satellite System (GNSS), such as the Global Positioning System (GPS), is regarded as a comprehensive and convenient tool, widely used to provide location data for ITS and in-vehicle applications. GNSS is widely applied in vehicle positioning through various methods [2], [3], [4], [5].

Advances in vehicular ad hoc networks, roadside units (RSUs), and ultrawideband (UWB) technology have driven significant progress in cooperative localization, enhancing vehicle positioning for ITS. Various schemes integrating GNSS information have been proposed to improve accuracy [6], [7], [8], including a recent method using an improved K-means++ algorithm for distributed localization [9]. This method integrates GPS information with intervehicle distance data and performs distributed localization after clustering the nodes.

The above methods all require GNSS coverage or assistance to estimate vehicle positions and cannot fully overcome the limitations in GNSS-denied environments. It is widely acknowledged that in complex environments, such as forests, urban canyons, tunnels, and multilevel bridges, the availability and reliability of GNSS signals are significantly reduced, and vehicles are often subject to interference from high-traffic volume and obstacles [10], [11], [12].

Many studies have proposed solutions to address these issues [13], [14], [15]. Hu et al. [16] proposed a method that utilizes kilometer signs to assist in correcting the errors

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of the inertial navigation system and odometer integration, aiming to improve the accuracy of autonomous vehicle navigation in a GNSS-denied environment. Wang et al. [17] proposed a map-based vehicle localization method, LiDAR, IMU, vision and wheel encoder data (LIVW) localization, that fuses LiDAR, vision, inertial measurement unit (IMU), and wheel encoder data to improve the accuracy and robustness in complex environments. Ouyang et al. [18] proposed a navigation mode combining dead reckoning, relative ranging, and heading constraint to address the cooperative localization of unmanned aerial vehicle (UAV) swarms in GNSS-denied environments. Tang et al. [19] proposed OdoNet, a 1-D convolutional neural network (CNN)-based pseudo-odometer model that learns from a single IMU to provide forward speed aiding without requiring hardware odometers. In addition, several methods based on other sensors have also been proposed [20], [21].

Ultrahigh frequency radio frequency identification (UHF-RFID) technology has garnered significant attention due to its low cost and suitability as a positioning solution in GNSS-denied environments [22], [23]. With the advancement of UHF-RFID technology, several positioning algorithms based on this technology have emerged. A vehicle positioning algorithm based on UHF-RFID has been proposed to address GNSS-denied environments [24]. This method installs an RFID reader on the vehicle, which communicates with roadside RFID tags using multifrequency signals to perform vehicle positioning. While this method performs well in positioning, it does not utilize the distance information between vehicles, and the deployment of roadside tags is constrained by environmental factors, especially when the roads are too wide.

In recent years, the UWB technology has made rapid advancements. This novel wireless communication technology offers high-ranging accuracy and several advantages, including low system complexity and a low probability of interception [25]. Recognized as a key future communication technology, UWB enables vehicle positioning systems for the Internet of Things (IoT) in transportation. Its integrated system, comprising base stations, tags, and a management platform, supports precise positioning in GNSS-denied environments like tunnels, forests, and indoor spaces [26], making it highly valuable for vehicle monitoring.

Inspired by this, the article proposes a scheme that integrates UWB technology with the vehicular network technology for localization in environments without GNSS coverage. The method involves setting up UWB base stations at regular intervals along both sides of the road. In dense traffic situations, UWB base stations may experience NLOS effects when localizing vehicles. To tackle this issue, we propose a strategy that begins by classifying vehicles into LOS and NLOS categories based on distance and AoA measurements, as well as discrepancies observed between two different localization algorithms. Then, the UWB base stations apply a specific positioning algorithm to locate LOS vehicles. Subsequently, based on the distance information between LOS and NLOS vehicles, measured by onboard devices such as laser radars, corresponding positioning methods are employed for NLOS-vehicle localization. Finally, by utilizing the estimated initial positions of all the vehicles and integrating the distance information between them, an iterative positioning algorithm is used to further refine the initial estimations.

The main contributions of this article can be summarized as follows.

First, this article presents an innovative strategy for cooperative vehicle localization in GNSS-denied environments. It introduces an efficient algorithm to classify vehicles into NLOS and LOS categories, along with algorithms specifically designed to calculate preliminary location estimates for each category.

Second, the localization of all the vehicles is further refined through the fusion of intervehicle distance information and preliminary positioning data, formulated as a standard quadratic programming model. By employing an efficient iterative positioning algorithm, the final localization results exhibit significantly enhanced accuracy.

Overall, these contributions position this article as a significant advancement in the field of vehicle localization in GNSS-denied environments, particularly in improving positioning accuracy in complex scenarios characterized by both LOS and NLOS conditions. The rest of this work is organized as follows. Section II models the problem of fusion positioning for vehicles within the selected road segment using roadside deployed UWB base stations and the measured distance information between vehicles in a GNSS-denied environment. Section III mainly introduces the specific positioning algorithm. In Section IV, we present numerical simulation experiments and results pertaining to this algorithm. Section V summarizes the contributions of this article and outlines directions for future work.

Throughout this article, the matrix $\mathbf{X}'$ denotes the transpose of matrix $\mathbf{X}$. For a square matrix $\mathbf{X}$, its inverse is denoted by $\mathbf{X}^{-1}$. $\mathbf{I}$ and $\mathbf{O}$ represent the identity and the zero matrix of appropriate dimensions, respectively. The symbol $||*||$ denotes the Euclidean norm.

II. Localization Scenario and Measurement Models

We assume a GNSS-denied vehicle positioning scenario where vehicles travel along the road. A road segment of length $L(m)$ and width $W(m)$ is selected, which includes n vehicles ($n > 3$). The topological structure formed by the vehicles on the road and the distances between them is represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, 2, 3, \ldots, n\}$ denotes the set of vehicle nodes and $\mathcal{E}$ denotes the set of edges between the vehicles. UWB base stations are deployed along the roadside at regular intervals, with the distance between adjacent base stations. In real-world scenarios, vehicles on the road transmit intervehicle distance information to the roadside-deployed UWB base stations. The UWB base stations measure the distances between themselves and each vehicle on the road while simultaneously receiving the intervehicle distance information transmitted by the vehicles. They then organize and analyze these data to estimate the positions of the vehicles. Finally, the position information are sent back to the vehicles. To ensure the full coverage of the road segment, such that each vehicle on the road is detected by at least two UWB base stations, the monitoring range of the base stations

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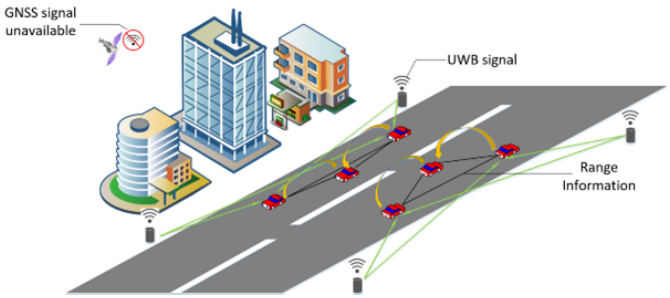


Fig. 1. Collaborative positioning scheme in GNSS-denied environment.

must satisfy the following condition in relation to the width of the road:

$$R > \sqrt{\frac{W^2 + d^2}{4}} \tag{1}$$

where R represents the fixed monitoring range of the UWB base stations, with $R(m)$ being the radius of the circular area, and $d(m)$ denotes the distance between two adjacent base stations. This article integrates UWB signals with onboard laser radar for vehicle positioning in a GNSS-denied environment. The specific fusion method is illustrated in Fig. 1.

The actual position of each vehicle is denoted by $\mathbf{x_i} = [x_i, y_i]'(i = 1, 2, 3, \ldots, n)$. The actual distance between vehicle node i and vehicle node j is expressed as follows:

$$d_{i,j} = ||\mathbf{x}_i - \mathbf{x}_j||. \tag{2}$$

For vehicles i and j, the measured distance typically contains errors compared to the actual distance. These errors usually follow a Gaussian distribution with zero mean. Therefore, the measured distance between vehicles i and j is represented by the following formula:

$$\hat{d}_{i,j} = d_{i,j} + \varepsilon_{i,j} \tag{3}$$

where $\varepsilon_{i,j}$ represents the measurement error, which follows a Gaussian distribution with zero mean (i.e., $\varepsilon_{i,j} \sim \mathbb{N}(0, \sigma^2)$, where σ^2 is the variance of the error).

It is important to note that the two noisy range measurements, $\hat{d}_{i,j}$ and $\hat{d}_{j,i}$ are collected by vehicle i and vehicle j, respectively. These measurements may be unequal due to varying levels of interference during the data collection process. Therefore, let n_i be the number of neighboring vehicles for vehicle i. The total number of intervehicle distances can be expressed as follows:

$$m = \sum_{i=1}^{n} n_i. \tag{4}$$

The actual distance vector between vehicles can be expressed as follows:

$$\mathbf{d} = \left[d_{1,1_1}, \ldots, d_{1,1_{n_1}}, d_{2,2_{n_2}}, \ldots, d_{2,2_{nn}}, \ldots, d_{n,n_{nn}}\right]' \in R^m \tag{5}$$

where d_{i,i_j} represents the actual distance between vehicle i and its jth neighboring vehicle. Then, the measured distance vector between vehicles can be expressed as follows:

$$\hat{\mathbf{d}} = \left[d_{1,1_1}, \ldots, d_{1,1_{n_1}}, d_{2,2_{n_2}}, \ldots, d_{2,2_{nn}}, \ldots, d_{n,n_{nn}}\right]' + \boldsymbol{\varepsilon} \tag{6}$$

where $\boldsymbol{\varepsilon}$ denotes the corresponding distance measurement noise vector with a covariance matrix $\mathbf{Q}_1$.

The measured distance between UWB base station i and vehicle j can be expressed as follows:

$$\hat{r}_{i,j} = r_{i,j} + \epsilon_{i,j} \tag{7}$$

where $r_{i,j}$ denotes the actual distance between UWB base station i and vehicle j. Then, the measured distance vector between UWB base stations and vehicles can be expressed as follows:

$$\hat{\mathbf{r}} = \left[r_{1,1}, r_{1,2}, r_{1,3}, \ldots, r_{k,n}\right]' + \boldsymbol{\epsilon} \tag{8}$$

where k is the total number of UWB base station and $\boldsymbol{\epsilon}$ represents the corresponding range measurement noise vector.

In a 2-D scenario, let $\mathbf{s} = [x_s, y_s]'$ be the coordinates of UWB base station j. We define the actual AoA of vehicle i to UWB base station j as the angle between the line connecting vehicle i and UWB base station j and the direction of the road, i.e., the positive direction of the x-axis, which can be expressed as follows:

$$\theta = \arccos \frac{x_i - x_s}{||\mathbf{x}_i - \mathbf{s}||}. \tag{9}$$

When the measurement noise is present, the measured value of AoAs can be expressed as follows:

$$\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + \boldsymbol{\eta} \tag{10}$$

where $\hat{\boldsymbol{\theta}} = [\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_n]'$ and $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_n]'$, $\boldsymbol{\eta}$ is a vector that follows the Gaussian distribution with zero mean.

III. Solution for GNSS-Denied Environment Localization

This article focuses on classifying vehicles into LOS and NLOS categories based on UWB base stations and collaboratively estimating vehicle positions by fusing the distance and AoA information obtained from UWB base stations, along with the distance information between vehicles. First, we categorize the positioning environment into two main scenarios based on whether the vehicles are affected by NLOS factors when utilizing UWB signals for localization.

1) Class I: All vehicles in the positioning environment are not affected by NLOS factors when utilizing UWB signals for localization.

2) Class II: Some vehicles on the road are affected by NLOS factors when utilizing UWB signals for localization, rendering them incapable of being directly located by roadside UWB base stations.

In the scenario described in Class I, not all vehicles are affected by NLOS factors; thus, it is possible to use the known locations of roadside UWB base stations and their distance information to locate the vehicles on the road. After obtaining the initial positioning results, an iterative optimization algorithm proposed in [31] is employed to fuse the

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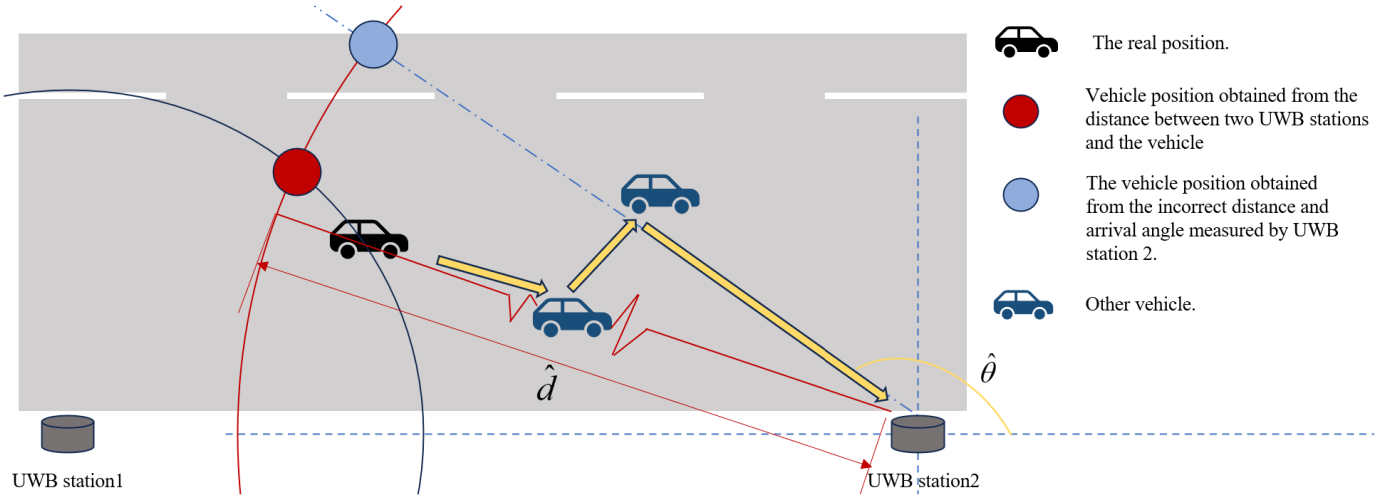


Fig. 2. Reason for the positioning deviation caused by the influence of NLOS on vehicles during distance measurement at UWB base stations.

preliminary positioning results with the distance information between vehicles, ultimately yielding more accurate position information for each vehicle.

In the environment described in Class II, we first need to classify the vehicles on the road, selecting the LOS vehicles that were not affected by NLOS factors during the UWB distance measurements. These LOS vehicles are then positioned using the coordinates of the UWB base stations and their distance information. Next, we locate the NLOS vehicles based on the known positions of the LOS vehicles and the distances between them and the unknown positions of the NLOS vehicles. Finally, we employ an iterative optimization algorithm to fuse the preliminary positions of the vehicles with the distance information between them, optimizing the initial positioning results to obtain the final position estimates for all vehicles on the road. For simplicity, we will denote Class I environment as the UWB-LOS environment and Class II environment as the UWB-NLOS environment in the following.

In this section, we will provide a detailed introduction to the proposed vehicle positioning algorithm for GNSS-denied environments from three aspects. First, we will define and categorize the LOS and NLOS vehicles within the selected road segment, analyzing their feasibility in both UWB-LOS and UWB-NLOS environments. Second, we will present the positioning method for the NLOS vehicles. Finally, we will introduce the algorithm for obtaining the final results through fusion iterative positioning, utilizing the preliminary positioning results and the distance relationships between vehicles.

A. Classification and Positioning of LOS Vehicles

First, when the UWB base stations deployed along the roadside receive signals transmitted from the UWB tags mounted on vehicles, they process the received signals to obtain the distance to the signal source and the AoA. At this point, to avoid interference from NLOS factors, a preliminary definition of the signal source is necessary to determine whether it belongs to LOS vehicles. Assuming the measurement distance to the signal source measured by the UWB base station is $\hat{l}$ and AoA is $\hat{\theta}$, the following formula can be used to determine whether it is sufficiently close to the roadside

$$\lambda_1 = \frac{\hat{l} \times \sin\left(\hat{\theta}\right)}{K_1}$$
$$f(\lambda_1) = \begin{cases} 1, & \lambda_1 \leq 1 \\ 0, & \lambda_1 > 1 \end{cases} \tag{11}$$

where K_1 is a value set manually, which is generally defined as 1/4 of the total width of the road. If $f(\lambda_1) = 1$, it indicates that the signal source vehicle is sufficiently close to the roadside; otherwise, it suggests that the signal source vehicle is farther from the roadside, which increases the likelihood of being affected by NLOS factors. However, this is not comprehensive, as even vehicles that are sufficiently close to the roadside may still be affected by NLOS factors during the signal reception process by the UWB base station if the distance between them is considerable or if there are partial obstructions between them. In addition, it is possible for vehicles that are farther from the roadside to not be affected by NLOS factors. Therefore, to avoid processing errors in the aforementioned two situations, it is necessary to use other methods to determine whether the vehicles will be affected by NLOS factors.

When performing UWB positioning, if NLOS factors are present, the measured distance and AoA will exhibit significant deviations. In this case, the circular positioning from two base stations and the AoA positioning from one base station will result in considerable differences, as illustrated in Fig. 2. In Fig. 2, UWB Station 2 is affected by NLOS factors, resulting in significant deviations in both distance measurement and AoA measurement. The red point represents the positioning result derived from AoA and distance measurement values affected by NLOS factors from UWB Station 2, while the blue point represents the positioning result obtained from the distance values of two UWB stations through geometric relationships.

This characteristic can be utilized to determine whether a vehicle is affected by NLOS, leading to the establishment of

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the following model:

$$\lambda_2 = \frac{||\hat{\mathbf{x}}_\rho - \hat{\mathbf{x}}_d||}{K_2}$$
$$g(\lambda_2) = \begin{cases} 1, & \lambda_2 \le 1 \\ -1, & \lambda_2 > 1 \end{cases} \tag{12}$$

where $\hat{\mathbf{x}}_\rho$ represents the positioning result derived from the AoA and distance measurement values affected by NLOS factors from the base station, which is the intersection point between the vector emitted in the direction of the AoA measured by the UWB base station and a circle centered at the base station, with a radius equal to the measured distance to the vehicle. $\hat{\mathbf{x}}_d$ represents the positioning result obtained from the distance measurements from two base stations which is the intersection point of two circles, each centered at one of the two base stations, with a radius equal to the distance between each base station and the vehicle. K_2 is a manually set parameter. In fact, (12) determines whether a vehicle is affected by NLOS factors by comparing the positioning results from two different localization methods. If $g(\lambda_2) = 1$, the signal source is considered to be unaffected by NLOS factors; if $g(\lambda_2) = -1$, it is considered to be affected by NLOS factors.

Combining with the model provided in (11), the overall model for determining whether a vehicle is an LOS vehicle is as follows:

$$\lambda = f\left(\frac{\hat{l} \times \sin\left(\hat{\theta}\right)}{K_1}\right) + g\left(\frac{||\hat{\mathbf{x}}_\rho - \hat{\mathbf{x}}_d||}{K_2}\right)$$
$$\mathrm{F}(\lambda) = \begin{cases} 1, & \lambda > 0 \\ 0, & \lambda \le 0. \end{cases} \tag{13}$$

If $\mathrm{F}(\lambda) = 1$, it indicates that the vehicle is classified as the LOS vehicle; otherwise, it is classified as the NLOS vehicle. The manually set K_1 and K_2 are used to control the sensitivity of the judgment. The settings of these parameters need to be adjusted based on the noise level in the actual environment and the severity of NLOS effects. Once the LOS vehicles have been classified, their positioning needs to be determined. Using geometric relationships, the intersection points of two circles, each centered at one of the two UWB base stations with radii equal to the distances from the vehicle to the respective base stations, can be taken as potential coordinates for the vehicle. Then, by applying prior knowledge to exclude unreasonable coordinates, an initial positioning result for the LOS vehicles can be obtained.

It is worth noting that when defining and initially positioning LOS vehicles, the density of vehicles prevents the simultaneous use of UWB base stations on both sides of the road for distance measurement on a single vehicle. This is because the UWB base stations on opposite sides may experience significant distance measurement errors due to the high number of vehicles, which can lead to deviations in the positioning results or even result in a complete failure to obtain a position. Therefore, it is generally advisable to use two or more UWB base stations on the same side of the sampled road to define and initially position LOS vehicles. The overall algorithm design is illustrated in Algorithm 1.

Algorithm 1 Classification and Positioning of LOS Vehicles

Input: $\mathbf{r}$, θ, Loc_{ap}, K_1, K_2
Output: $\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_1', \tilde{\mathbf{x}}_2', \ldots\ldots \tilde{\mathbf{x}}_n']'$

while $i<n$ **do**
  Inspection step1.Judge if the current vehicle is close enough to the roadside. Calculate:

$$x_1 = \frac{r_{i,1} \times sin(\theta_{i,1})}{K_1}$$
$$x_2 = \frac{r_{i,2} \times sin(\theta_{i,2})}{K_1}$$
$$\lambda_{1i} = max(x_1, x_2)$$

  if $\lambda_{1i} <= 1$ **then**
    $f(\lambda_{1i}) = 1$
  else
    $f(\lambda_{1i}) = 0$
  end
  Here, $r_{i,j}$ and $\theta_{i,j}$ represent the distances and arrival angle between the i-th vehicle and j-th UWB base station respectively.
  Instection step2. Judge whether the current vehicle affected by NLOS.
  (1) Calculate: $\tilde{\mathbf{x}}_{d_i}$.
  (2) Calculate: $\tilde{\mathbf{x}}_{\rho_i}$.
  (3) Following the (12)., we calculate:

$$\lambda_{2i} = \frac{||\tilde{\mathbf{x}}_{\rho_i} - \tilde{\mathbf{x}}_{d_i}||}{K_2}$$

  if $\lambda_{1i} <= 1$ **then**
    $g(\lambda_{2i}) = 1$
  else
    $g(\lambda_{2i}) = -1$
  end
  if $f(\lambda_{1i}) + g(\lambda_{2i}) > 0$ **then**
    $\tilde{\mathbf{x}}_i = \tilde{\mathbf{x}}_{\rho_i}$ or $\tilde{\mathbf{x}}_{d_i}$
  else
    $\tilde{\mathbf{x}}_i = null$
  end
  $i = i + 1$
end

In the UWB-LOS scenario, the algorithm will ultimately label all vehicles on the road as LOS vehicles and perform initial positioning for them. However, in the UWB-NLOS scenario, the algorithm will identify and label the LOS vehicles for initial positioning, while marking the NLOS vehicles on the road as unknown vehicles. These unknown vehicles will then be carried forward to the next step of the algorithm for grouping and initial positioning.

B. Grouping and Positioning of NLOS Vehicles

Once the preliminary coordinates of the LOS vehicles are calculated, the vehicles with unknown positions are grouped based on the known coordinates of the vehicles and the distance relationships between them and the unknown vehicles.

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Before grouping the NLOS vehicles, it is crucial to first group the LOS vehicles that have been initially located using the UWB base stations. Along the direction of the road, every v adjacent LOS vehicles are grouped together until the number of ungrouped LOS vehicles becomes an integer v_{min}, at which point the last v_{min} vehicles are grouped together. Since the grouping process is conducted along the road direction, the final v_{min} vehicles must be adjacent vehicles.

For the grouping of LOS vehicles, adjacent LOS vehicles are grouped along the direction of the road. This approach ensures spatial proximity within each group and aligns with the linear distribution characteristics of vehicle queues along the roadway, thereby reducing the cross-lane signal interference. Moreover, vehicles within each group must satisfy the LOS determination condition as defined in (13). The number of vehicles v in each group should be dynamically adjusted based on road traffic density, with a default value of $v = 3$. This group size not only facilitates the application of multilateration and the least squares method for positioning NLOS vehicles but also helps to manage computational complexity.

Using the above algorithm, the known positions of LOS vehicles can be grouped, which will serve as the basis for further classification of unknown-position vehicles. For each NLOS vehicle, the algorithm will calculate its distance to each existing group and classify it into the nearest group. It will then use the positions of the vehicles in that group, along with the distance relationships between them and the unknown NLOS vehicle, to estimate the position of the NLOS vehicle. The definition of the distance between the vehicle and each group is as follows.

Definition 3: Let $\mathcal{S} = \{S_1, S_2, \ldots, S_k\}$ be the set of already classified groups. If the current NLOS vehicle i has an unknown position, the distance from vehicle i to group S_s is given by

$$D_{i,s} = \frac{\min(L_1, L_2)^2}{L_1 + L_2} + \frac{\max(L_1, L_2)^2}{L_1 + L_2} \tag{14}$$

where L_1 and L_2 is expressed as follows:

$$\begin{aligned} L_1 &= \min_{D_{i,s}} \left\{ \hat{d}_{i,j} \right\}, \quad j = 1, 2, \ldots, k \\ L_2 &= \min_{D_{i,s}} \left\{ \hat{d}_{i,j} \right\}, \quad k = 1, 2, \ldots, j-1, j+1, \ldots, k \end{aligned} \tag{15}$$

where $\hat{d}_{i,j}$ is the measured distance between the unknown vehicle i and vehicle j in group S_s.

We use the least squares method to calculate the position of the NLOS vehicle [30]. For the unknown vehicle i, if the distances to the m vehicles in group S_s, with known coordinates (x_j, y_j), are represented as $\{d_{i,1}, d_{i,2}, \ldots, d_{i,m}\}$, the relationship between its position and the distances to the vehicles can be represented by the following model:

$$\begin{cases} \sqrt{(x_i - x_1)^2 + (y_i - y_1)^2} = d_{i,1} + \varepsilon_1 \\ \sqrt{(x_i - x_2)^2 + (y_i - y_2)^2} = d_{i,2} + \varepsilon_2 \\ \ldots \\ \sqrt{(x_i - x_m)^2 + (y_i - y_m)^2} = d_{i,m} + \varepsilon_m \end{cases} \tag{16}$$

where ε_i represents the Gaussian noise with a mean of 0. By applying the least squares method to solve this equation, the estimated position $\hat{\mathbf{x}} = (x_i, y_i)$ of the unknown vehicle can be obtained. The analytical solution for its final position is shown in (15)

$$\hat{\mathbf{x}} = (\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{B} \tag{17}$$

where

$$\mathbf{A} \triangleq 2 \times \begin{bmatrix} (x_1 - x_m) & (y_1 - y_m) \\ (x_2 - x_m) & (y_2 - y_m) \\ \cdots & \\ (x_{m-1} - x_m) & (y_{m-1} - y_m) \end{bmatrix} \tag{18}$$

and

$$\mathbf{B} \triangleq \begin{bmatrix} x_1^2 - x_m^2 + y_1^2 - y_m^2 + d_{i,m}^2 - d_{i,1}^2 \\ x_2^2 - x_m^2 + y_2^2 - y_m^2 + d_{i,m}^2 - d_{i,2}^2 \\ \cdots \\ x_{m-1}^2 - x_m^2 + y_{m-1}^2 - y_m^2 + d_{i,m}^2 - d_{i,m-1}^2 \end{bmatrix} \tag{19}$$

and $d_{i,j}$ denotes the distance between unknown vehicle i and known vehicle j in group S_s.

For the grouping and positioning of NLOS vehicles, the overall algorithm design is shown in Algorithm 2.

This algorithm does not require consideration of the spatial distribution of these vehicles when grouping and initially positioning NLOS vehicles. In other words, the algorithm can theoretically group and initially position all NLOS vehicles regardless of their distribution. In addition, when calculating the distance between NLOS vehicles and each regional group, the algorithm incorporates weights, so that the influence of closer vehicle nodes on the distance calculation result is greater than that of farther vehicle nodes. This helps reduce the potential impact of NLOS factors.

C. Iterative Optimization of Preliminary Estimates

After the preliminary positions of all vehicles within the selected road segment have been computed using the aforementioned two steps, they will be integrated with the intervehicle distance information for further refinement. Due to the errors present in the preliminary estimation of vehicle positions, we use $\hat{\mathbf{x}}_i$ to represent the initial positioning results, which is expressed as follows:

$$\hat{\mathbf{x}}_i = \mathbf{x}_i + \mu_i \tag{20}$$

where μ_i is the zero-mean Gaussian noise. Then, the vector of initial positioning results can be expressed as follows:

$$\hat{\mathbf{x}} = \left[\mathbf{x}_1', \mathbf{x}_2', \ldots, \mathbf{x}_n'\right]' + \boldsymbol{\mu} \tag{21}$$

where $\boldsymbol{\mu}$ denotes the corresponding noise vector with covariance matrix $\mathbf{Q}_2$.

Due to the Gaussian noise assumption, by combining the initial vehicle position estimates with the intervehicle distance information and constructing the estimation vector $\hat{\mathbf{u}} \triangleq [\hat{\mathbf{x}}', \hat{\mathbf{d}}']'$, the likelihood function with respect to the position vector x can be formulated as follows:

$$p(\mathbf{x}|\hat{\mathbf{u}}) = \frac{1}{(2\pi)^{\frac{m}{2}}\sqrt{|\mathbf{Q}_2|}} \times \mathbf{e}^{-\frac{1}{2}(\hat{\mathbf{x}}-\mathbf{x})'\mathbf{Q}_2^{-1}(\hat{\mathbf{x}}-\mathbf{x})}$$

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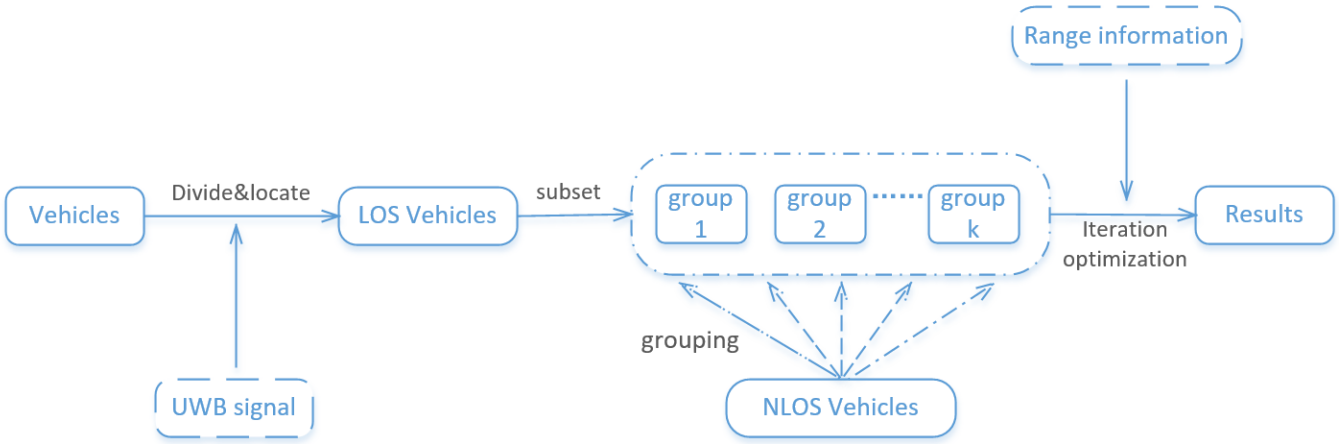


Fig. 3. Overall positioning process of vehicles in NLOS environment.

Algorithm 2 Grouping and Positioning of NLOS Vehicles

Input: $\tilde{\mathbf{x}}$, $\hat{\mathbf{d}} \in R^{n\times m}$
Output: $\hat{\mathbf{x}} \triangleq [\hat{\mathbf{x}}_1', \hat{\mathbf{x}}_2', \ldots, \hat{\mathbf{x}}_n']'$
Data: The number of LOS-vehicles C_{LOS}, the low bound of the remaining LOS-vehicles v_{min}, the number of unknown vehicles $C_{unknown}$.

begin
  $Count \leftarrow C_{LOS}$;
  $N_{team} \leftarrow 1$;
  $c \leftarrow 1$
  while $Count > v_{min}$ **do**
    1) Find the LOS-vehicle on the far left that is not grouped, denoted as $V_{current}$;
    2) $V_{current} \rightarrow C_{N_{team}}$;
    3) $c \leftarrow c + 1$;
    4) if $c > v_{min}$ then $c = 1$, $N_{team} \leftarrow N_{team} + 1$;
    5) $Count = Count - 1$.
  end
  Divide the remaining v_{min} vehicles into a group.
  After the above process, we can obtain the groups of LOS-vehicles marked $C_1, C_2, \ldots, C_k$. **while** $C_{unknown} > 0$ **do**
    Select the vehicles that has not been located, marked $v_{unknown}$.
    Set: $dist_{min} = inf$, $part_{min} = 1$, $i = 1$. **while** $i \leq k$ **do**
      Calculate the $D_{i,s}$ according to (14) and (15).
      $D_{i,s} = (min(L_1, L_2)/L_1+L_2)^2+(max(L_1, L_2)/L_1+L_2)^2$
      if $D_{i,s} < dist_{min}$ **then**
        $dist_{min} \leftarrow D_{i,s}$;
        $v_{unknown} \rightarrow C_i$;
        Calculate the location of $v_{unknown}$ according to (17).
        $\hat{\mathbf{x}} = (\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{B}$
      end
      $i \leftarrow i + 1$;
    end
  end
end

$$\times \frac{1}{(2\pi)^{\frac{m}{2}}\sqrt{|\mathbf{Q}_1|}} \times \mathbf{e}^{-\frac{1}{2}\left(\hat{\mathbf{d}}-\mathbf{d}\right)'\mathbf{Q}_1^{-1}\left(\hat{\mathbf{d}}-\mathbf{d}\right)}. \quad (22)$$

where $\mathbf{Q}_1$ and $\mathbf{Q}_1$ represent the noise covariance matrices. Using the maximum likelihood principle, we formulate the positioning problem, which leads to the following optimization problem with respect to the position vector $\mathbf{x}$:

$$\hat{\mathbf{x}}_{ML} = \arg\min_{\mathbf{x}} \frac{1}{2}\left(\hat{\mathbf{x}}-\mathbf{x}\right)'\mathbf{Q}_2^{-1}\left(\hat{\mathbf{x}}-\mathbf{x}\right) + \frac{1}{2}\left(\hat{\mathbf{d}}-\mathbf{d}\right)'\mathbf{Q}_1^{-1}\left(\hat{\mathbf{d}}-\mathbf{d}\right). \quad (23)$$

To solve this optimization problem, we introduce a new variable vector $\mathbf{u} \triangleq [\mathbf{x}', \mathbf{d}]'$ and regard $d_{i,j} = ||\mathbf{x}_i - \mathbf{x}_j||$ as the constraint condition. Then, we can formulate the equivalent problem as follows:

$$\min_{\mathbf{x}} \mathbf{u}'\mathbf{C}\mathbf{u} - 2\hat{\mathbf{u}}'\mathbf{C}\mathbf{u}$$
$$\text{s.t. } f_{ij}(\mathbf{u}) = 0, \quad \forall (i,j) \in \mathcal{V} \quad (24)$$

where $\mathbf{C} \triangleq [\mathbf{Q}_2 \quad \mathbf{O}$
$\mathbf{O} \quad \mathbf{Q}_1]$, $f_{i,j}(\mathbf{u}) \triangleq (\mathbf{x}_i - \mathbf{x}_j)'(\mathbf{x}_i - \mathbf{x}_j) - d_{ij}$. After reformulation, the original problem is expressed as a quadratically constrained quadratic programming (QCQP) problem, which is nonconvex in nature. It is well known that such problems are generally NP-hard [27], and only certain special cases of nonconvex QCQP problems can be solved in the polynomial time [28], [29]. To address this problem, we decompose problem (22) into a series of simpler subproblems. We replace the nonconvex quadratic constraints with some appropriate convex approximations. We first assume that an interpolated estimate of u is available. In this problem, we use the initial vehicle positions $\hat{\mathbf{x}}_i$, calculated from the previous two steps, as the initial estimate, while the measured distances $\hat{\mathbf{d}}_{i,j}$ between them can be regarded as the initial estimate for $\mathbf{d}_{i,j}$. Thus, we replace $(\mathbf{x}_i - \mathbf{x}_j)'$ with $(\hat{\mathbf{x}}_i - \hat{\mathbf{x}}_j)'$ and i, j with $\hat{d}_{i,j}$, to define the following approximation for $f_{i,j}(\mathbf{u})$:

$$\tilde{f}_{i,j}(\mathbf{u}; \hat{\mathbf{u}}) = 2(\hat{\mathbf{x}}_i - \hat{\mathbf{x}}_j)'(\mathbf{x}_i - \mathbf{x}_j) - 2\hat{d}_{i,j}d_{i,j}. \quad (25)$$

Function $\tilde{f}_{i,j}$ can be regard as the convex approximation of $f_{i,j}$ at the point $\hat{\mathbf{u}}$. Based on this approximation, the original nonconvex problem (20) can be approximated as the following

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Algorithm 3 Iterative Localization Algorithm

Input: $\hat{\mathbf{x}}', \hat{\mathbf{d}}, \mathbf{Q}_1, \mathbf{Q}_2, \gamma^k$
Output: $\hat{\mathbf{X}}' = [\hat{\mathbf{X}}_1'; \hat{\mathbf{X}}_2'; \ldots; \hat{\mathbf{X}}_n']'$

begin

1) Initialize $k = 0$, let $\tilde{\mathbf{u}}^k = [\hat{\mathbf{x}}', \hat{\mathbf{d}}']$.
2) Set $k = k + 1$. Update $\mathbf{A}^{k-1} = [a_{11_1}^{k-1}; a_{11_2}^{k-1}; \ldots; a_{11_{n_1}}^{k-1}; a_{22_1}^{k-1}; \ldots; a_{nn_{n_n}}^{k-1}]$, where $a_{ij}^{k-1} = (\hat{\mathbf{u}}^{k-1})'\mathbf{C}_{ij}$. The matrix $\mathbf{C}_{ij}$ is expressed as:
$\mathbf{C}_{ij}(2i-1:2i, 2i-1:2i) = \mathbf{C}_{ij}(2j-1:2j, 2j-1:2j) = \mathbf{I}_2$
$\mathbf{C}_{ij}(2i-1:2i, 2j-1:2j) = \mathbf{C}_{ij}(2j-1:2j, 2i-1:2i) = -\mathbf{I}_2$
$\mathbf{C}_{ij}(2n + \sum_{l=1}^{i-1} n_l + j, 2n + \sum_{l=1}^{i-1} n_l + j) = -1$
and all other elements in $\mathbf{C}_{ij}$ are equal to 0.
Then calculate $\mathbf{P}^k = I - \mathbf{A}^{k'}(\mathbf{A}^k\mathbf{A}^{k'})^{-1}\mathbf{A}^k$.
3) Solve the approximate problem (24), calculate the approximate solution:

$$\bar{\mathbf{u}} = (\mathbf{P}^{k-1}\mathbf{C}\mathbf{P}^{k-1})^{-1}\mathbf{C}\hat{\mathbf{u}}$$

4) Calculate the update of the k-th iteration as:

$$\tilde{\mathbf{u}}^k = \tilde{\mathbf{u}}^{k-1} + \gamma^{k-1}(\bar{\mathbf{u}} - \tilde{\mathbf{u}}^{k-1})$$

5) If:

$$\frac{||\tilde{\mathbf{u}}^k - \tilde{\mathbf{u}}^{k-1}||}{||\tilde{\mathbf{u}}^k||} \leq \delta$$

where $\delta > 0$ is a given small parameter to indicate the termination. Then the convergence is reached and the final output is $\tilde{\mathbf{u}}^k$, otherwise go to the step2.

end

convex form:

$$\min_{\mathbf{x}} \mathbf{u}'\mathbf{C}\mathbf{u} - 2\hat{\mathbf{u}}'\mathbf{C}\mathbf{u}$$
$$\text{s.t. } f_{i,j}(\mathbf{u}; \hat{\mathbf{u}}) = 0, \quad \forall (i,j) \in \mathcal{V}. \tag{26}$$

For problem (26), the equality constraints are clearly convex linear, and the objective function is quadratic; therefore, this problem is a standard quadratic programming problem with convex linear equality constraints. The greatest advantage of this approximation is that it has an explicit analytical solution. We utilize the iterative optimization algorithm proposed in [31] to compute the analytical solution for problem (26). The algorithm design is shown in Algorithm 3.

Through the three stages of defining and grouping the LOS vehicles, classifying and preliminarily locating the NLOS vehicles, and performing iterative optimization using the initial vehicle positions and the intervehicle distance information, we can obtain the final positions of all vehicles on the surveyed road segment.

D. Overall Positioning Process

The overall process for vehicle positioning in GNSS-denied environments can be summarized as follows based on the three positioning steps outlined above. First, LOS vehicles are classified using the distance and AoA information measured by the UWB base stations, and their initial positioning is performed. Second, based on the known positions of LOS vehicles and the intervehicle distance information, the NLOS vehicles are grouped and positioned. Finally, an iterative optimization algorithm is employed to integrate the initially estimated vehicle positions with the measured intervehicle distance information, resulting in the determination of the final positions of all vehicles. The overall process is illustrated in Fig. 3.

IV. Simulation Experiment

In this section, we will assess the performance of the proposed algorithm through two experiments. To ensure the accuracy and rigor of these experiments, the dimensions of the selected road segment will be fixed at a length of 50 m and a width of 40 m. However, in practical situations, vehicles located in the middle of the road can obstruct signals, making the UWB base stations on opposite sides of the road susceptible to NLOS factors. As a result, the data provided by these opposite-side UWB base stations may not be highly reliable. Therefore, in these experiments, we will primarily focus on the UWB base stations located on the same side of the road, with the width of the road set to half of the total width and considering only one direction of the roadway. The coordinates of the two UWB base stations are $\mathbf{X}_1 = [0, 0]'$ and $\mathbf{X}_2 = [50, 0]'$. In the experiment, we set the variance of the distance measurement noise between vehicles as σ_1 and the variance of the distance measurement noise between the UWB base stations and the vehicles as σ_2. The covariance matrices are defined as $\mathbf{Q}_1 = \sigma_1^2\mathbf{I}_{m\times m}$ and $\mathbf{Q}_2 = \sigma_2^2\mathbf{I}_{2n\times 2n}$. We set $K_1 = 10$ and set K_2 in the classification algorithm as $K_2 = 5 * \sigma_2^2$. For the iterative optimization algorithm, we initialized the update step size at $\gamma = 0.5$.

The root mean square error (RMSE) is a commonly used metric for measuring the discrepancy between the locations obtained by a positioning algorithm and their true positions. RMSE is particularly sensitive to outliers, making it an effective indicator of the performance of the positioning algorithm. The calculation method is provided by the following equation:

$$\text{RMSE}_x = \sqrt{\frac{\sum_{t=1}^{T}\sum_{i=1}^{n}\left(\hat{x}_i^t - x_i\right)^2}{nT}}$$
$$\text{RMSE}_y = \sqrt{\frac{\sum_{t=1}^{T}\sum_{i=1}^{n}\left(\hat{y}_i^t - y_i\right)^2}{nT}} \tag{27}$$

where $T = 1000$ is the total number of repetitions for each simulation experiment and n is the number of vehicles.

A. Simulation 1

We first investigate the classification performance of the proposed classification method. In this experiment, we fix the noise level of the ranging between the UWB base station and the vehicle at $\sigma_2 = 1.0$ and vary the number of vehicles from $n = 10$ to $n = 30$ to explore the relationship between classification correct rate and the number of vehicles. For each

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TABLE I
RELATIONSHIP BETWEEN ζ AND THE NUMBER OF VEHICLES

Number of vehicles	Average correct rate(%)	Number of vehicles	Average correct rate(%)
10	93.91%	21	97.67%
11	94.72%	22	97.26%
12	96.26%	23	95.93%
13	97.84%	24	97.19%
14	96.01%	25	96.67%
15	**98.32%**	26	**93.42%**
16	97.09%	27	95.74%
17	98.17%	28	96.11%
18	96.83%	29	96.86%
19	96.6%	30	96.75%
20	96.03%		

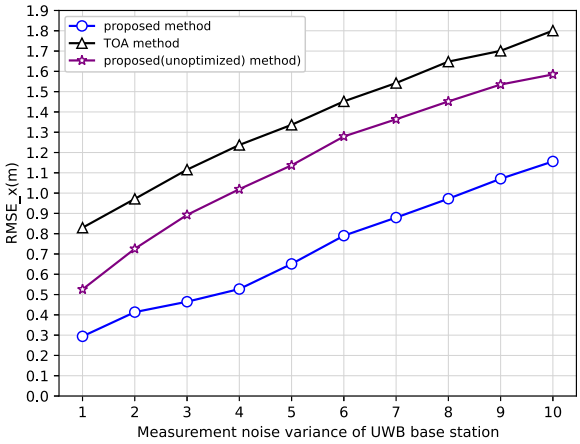


Fig. 4. Trend of RMSE$_x$ variation with variance of UWB measurement noise.

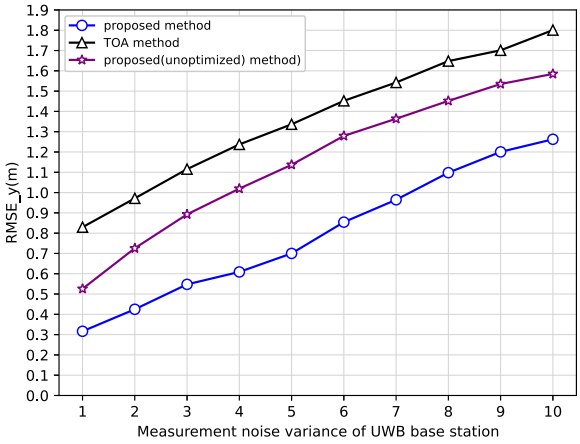


Fig. 5. Trend of RMSE$_y$ variation with the variance of UWB measurement noise.

experiment with n vehicles, let $T = 1000$ be the number of repetitions for that group of experiments. The positions of the n vehicles are randomly generated, and the NLOS vehicles, which are obstructed and are manually labeled. The noise variance of the measured distance between these NLOS vehicles and the UWB base station is set to $\hat{\sigma} = k_1 \times \sigma_2$, where $k_1 = 50$. Let $\mathbf{v} = [v_1, v_2, \ldots, v_n]'$ be the class vector of the vehicles, where $v_i = 0$ indicates that vehicle i is the NLOS vehicle, and $v_i = 1$ indicates that vehicle i is the LOS vehicle. Correspondingly, $\hat{\mathbf{v}} = [\hat{v}_1, \hat{v}_2, \ldots, \hat{v}_n]'$ represents the

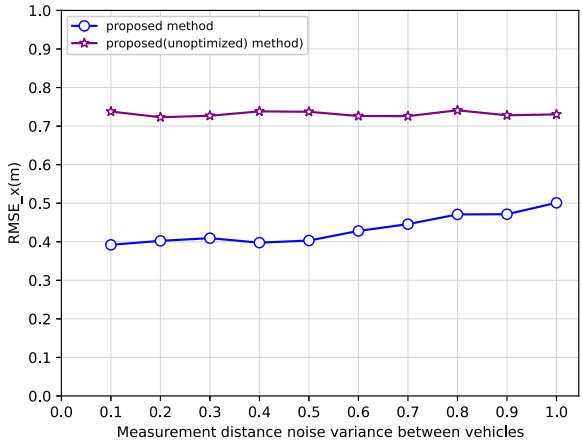


Fig. 6. Trend of RMSE$_x$ with variance of intervehicles distance measurement noise.

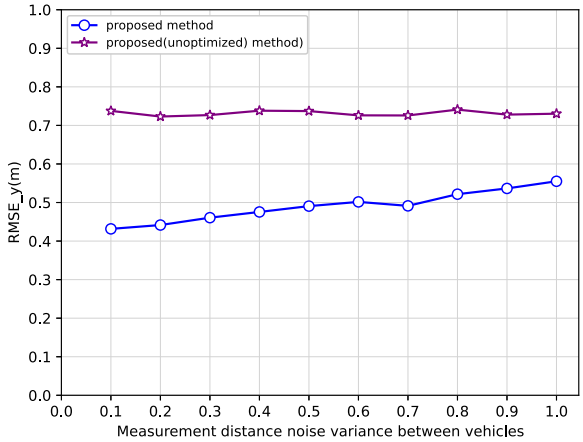


Fig. 7. Trend of RMSE$_y$ with variance of intervehicles distance measurement noise.

class vector obtained by the classification method. The correct rate ζ of the results obtained by the classification method can be expressed by the following formula:

$$\zeta = \frac{\sum_{t=1}^{T}\sum_{i=1}^{n}\left|v_i^t - \hat{v}_i^t\right|}{nT} \times 100\% \tag{28}$$

where n is the total number of the vehicles and $|v_i^t - \hat{v}_i^t|$ represents the absolute value of $v_i^t - \hat{v}_i^t$.

Table I shows the relationship between the correct rate of the classification method and the number of vehicles. From the results, it can be seen that as the number of vehicles increases from $n = 10$ to $n = 30$, the classification correct rate is not significantly affected, with the average correct rate remaining around 95%. From Table I, the minimum correct rate of the classification method is 93.42%, and the maximum correct rate is 98.32%. These results fully demonstrate the effectiveness of the algorithm in distinguishing between NLOS and LOS vehicles.

B. Simulation 2

In this section, we explore the relationship between the localization performance of the proposed algorithm and the measurement noise variance.

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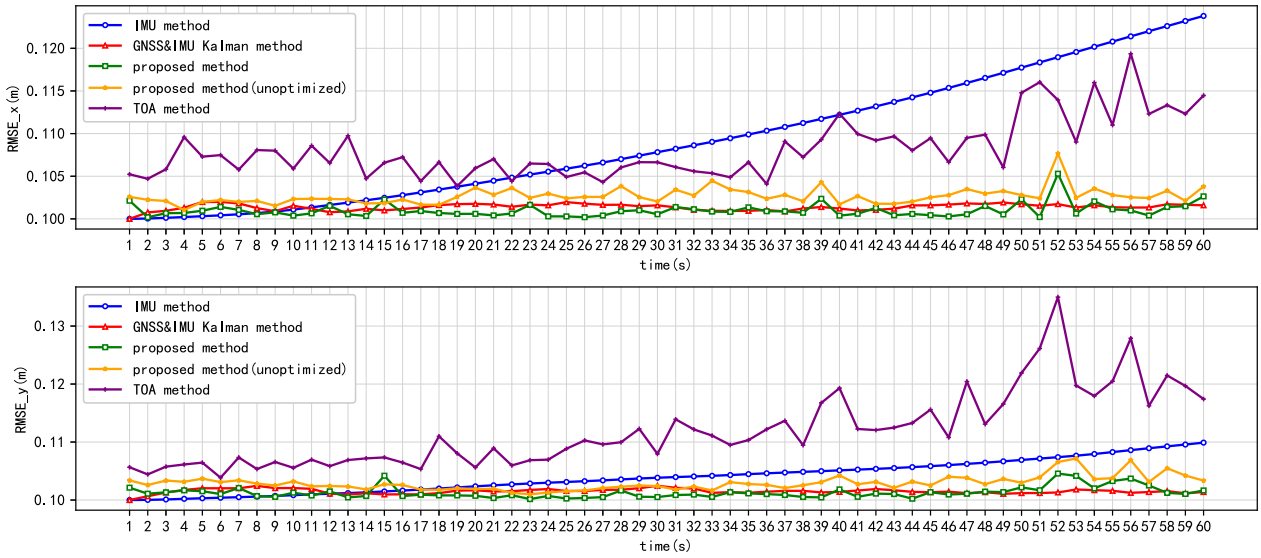


Fig. 8. Trend of RMSE_x and RMSE_y of various algorithms over time in dynamic scenarios.

First, we fixed the measurement noise variance for the distance between vehicles at $\sigma_1 = 0.1$ and the number of vehicles at $n = 10$. We investigated the relationship between RMSE in localization and the measurement noise variance σ_2 of the UWB base stations, comparing it with traditional TOA positioning methods. It is worth mentioning that during the TOA positioning process, we added a new UWB base station on the opposite side of the road, located at $X_3 = [0, 40]'$. This base station will work alongside the existing two UWB base stations to perform multilateration for the vehicles on the road. Figs. 4 and 5 present the simulation results as the measurement noise variance of the UWB base stations changes from $\sigma_2 = 1.0$ to $\sigma_2 = 10.0$.

Figs. 4 and 5 illustrate the variation trends of RMSE_x and RMSE, respectively, with respect to the measurement noise variance. As shown in Fig. 4, when the UWB ranging noise variance σ_2^2 increases from 1.0 to 10.0, the RMSE_x of the conventional TOA method rises from 0.82 to 1.79 m, whereas the RMSE_x of the proposed method without optimization increases only from 0.51 to 1.59 m. After iterative optimization, RMSE_x is further reduced to a range of 0.3–1.16 m. Compared with the conventional TOA-based localization, the proposed scheme achieves an average improvement of approximately 48%. Similarly, as presented in Fig. 5, the proposed localization method achieves an average improvement of about 41%. These results demonstrate that the proposed approach effectively reduces localization errors.

Second, we investigated the relationship between the localization performance of the proposed algorithm and the variance of the distance measurement noise between vehicles, denoted by σ_1. We also fixed the number of vehicles at $n = 10$ and the variance of UWB measurement noise at $\sigma_2^2 = 2.0$. The variance of the distance measurement noise between vehicles was varied from $\sigma_1^2 = 0.1$ to $\sigma_1^2 = 1.0$, based on the selected road segment. The aim was to observe the trends of various positioning algorithms. The results are presented in Figs. 6 and 7.

Analysis of the simulation data in Figs. 6 and 7 demonstrates that the proposed optimization algorithm consistently achieves significant performance improvements under various levels of intervehicle ranging noise ($\sigma_1^2 = 0.1 \sim 1.0$). Specifically, when $\sigma_1^2 = 0.1$, the optimized algorithm achieves a longitudinal localization error (RMSE_x) of 0.39 m, representing a 45.9% reduction compared with the unoptimized algorithm, and a lateral error (RMSE_y) of 0.42m, reflecting the 41.6% reduction. As the noise level increases to $\sigma_1^2 = 1.0$, the optimized RMSE_x and RMSE_y are maintained at 0.5 and 0.54 m, respectively, still yielding advantages of 29.6% and 23.9%.

These results highlight the superior performance of the proposed method compared to other approaches, demonstrating its effectiveness in achieving relatively accurate positioning in GNSS-denied environments.

C. Simulation 3

It is important to note that in real-world applications, vehicles are not stationary; on the contrary, they are typically in constant motion. Therefore, in this experiment, we will evaluate the tracking performance of the proposed method for moving vehicles. To effectively assess this, we introduce a trajectory-tracking method based on the IMU data as a comparison in addition to other algorithms.

In this simulation, we keep $\sigma_1^2 = 0.1$ and $\sigma_2^2 = 1.0$, the number of vehicles is fixed at $n = 10$, with the road section measuring 50 m in length and 40 m in width. The initial positions of the ten vehicles within the designated road section are randomly generated, along with their acceleration profiles over a time period of $t = 60$ s. It is noteworthy that, in this simulation, we only focus on vehicle positioning on one side of the road, and therefore, the velocity direction along the x-axis is always positive.

Fig. 8 illustrates the trend of the RMSE for vehicles along the x-axis and y-axis over a 60-s tracking period. As observed

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in the figure, the traditional TOA positioning algorithm suffers significantly from NLOS effects during dynamic positioning, resulting in substantial positioning errors. The proposed scheme effectively maintains both RMSE_x and RMSE_y within the range of 0.1 to 0.11 m in dynamic scenarios. This indicates that under conditions of $\sigma_1^2 = 0.1$ and $\sigma_2^2 = 1.0$, the proposed method can achieve localization performance comparable with that of a Kalman filter using GNSS observations with a noise level of $\text{sigma}_{\text{GNSS}} = 5.0$, while also effectively preventing error accumulation. In addition, it demonstrates tracking performance comparable with that of the Kalman filter-based IMU trajectory estimation. Overall, the proposed method is applicable in dynamic scenarios, where vehicles are continuously moving and exhibits the excellent tracking performance.

V. Conclusion

In this article, we present a novel approach for vehicle cooperative localization in GNSS-denied environments by harnessing roadside UWB base stations alongside relative distance information among vehicles. To tackle the challenge of initial position estimation for vehicles in these scenarios and to mitigate the adverse effects of NLOS factors on localization accuracy, we first classify vehicles into NLOS and LOS categories and design algorithms specifically to calculate preliminary location estimates for each category. We further refine these preliminary estimates through a maximum likelihood optimization strategy, implementing an iterative algorithm to derive final position estimates. Simulation experiments reveal that in static localization environments, our proposed method significantly outperforms traditional time-of-arrival localization algorithms, achieving lower positioning errors. Furthermore, in dynamic localization scenarios, our method effectively mitigates error accumulation, demonstrating superior performance in vehicle tracking.

The primary focus of this article is to provide a computational method for accurately estimating vehicle positions in GNSS-denied environments. Therefore, in our future work, enhancing the robustness and scalability of the proposed method will be a key research direction. For instance, we will optimize the proposed approach to better accommodate real-world scenarios where relative distance measurements between a few vehicle nodes may fail. Another significant avenue for future research is to reduce the computational time consumed during the localization process, thereby improving the timeliness of the proposed algorithm, making it more suitable for real-time dynamic tracking scenarios.

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